

TransGANomaly: Transformer based Generative Adversarial Network for Video Anomaly Detection

Nazia Aslam, Maheshkumar H Kolekar

*Video Surveillance Lab,
Department of Electrical Engineering,
Indian Institute of Technology Patna,
Bihta-801106, India*

Abstract

Video anomaly detection aims to identify a set of abnormal events in videos. Deep reconstruction and prediction-based models have been employed to detect anomalies. Deep reconstruction models sometimes recreate the abnormal events along with the normal ones. However, the prediction-based approaches have demonstrated encouraging results. This paper presents a video vision transformer (ViViT) based generative adversarial network (GAN), TransGANomaly, a novel approach for detecting anomalies. The proposed framework is a video frame predictor and trained only on normal video data adversarially. The generator of the GAN is a ViViT network that receives 3D input tokens from the video snippets. The generator aims to predict the future frame based on past sequences. After that, the predicted and original frames are given to the model's discriminator for binary classification. Extensive experiments have been performed on UCSD Pedestrian, CUHK Avenue, and ShanghaiTech datasets to validate the efficacy of the proposed method.

Keywords: Anomaly detection, Video Vision Transformer, Adversarial training, Generative Adversarial Network

1. Introduction

Video anomaly detection (VAD) is a predominant task in computer vision, which is employed in several essential applications, including video surveillance [47], [20], scene understanding [75], [63], [62], and abnormal human activity recognition [45], [19]. Frame-level VAD aims to recognize those frames which deviate from the regular ones for given video footage. In recent years, industry and academia have focused on developing robust video anomaly detection systems [45]. However, detecting anomalous events in videos still needs to be improved due to two challenges. 1) The definition of anomalies is highly ambiguous, and its pattern varies wildly depending on the applications and settings. For example, walking in the pedestrian pathway is a regular event, while walking in the middle of traffic is an abnormal event. Therefore, those VAD approaches which are resistant to varied conditions and environments are in high demand. 2) The data imbalance problem between the normal events and abnormal events. In day-to-day life, regular events are more likely than irregular ones. Therefore, collecting, annotating and classifying anomalous events are time-consuming and expensive in terms of money and people. As a result, the VAD algorithms are expected to learn from the normal data and differentiate the unseen and unbounded aberrant occurrences from normal ones. The deep learning-based VAD approaches are generally categorized in two ways:

i) reconstruction-based approaches and ii) prediction-based approaches. In reconstruction-based approaches, the model learns to recreate the input data with minimal error [4], [5]. Moreover, during testing, the higher reconstruction error of the pixel signifies an anomalous event. In contrast, the prediction-based approach learns to predict the future frame of input video data based on past sequences [6]. Moreover, the higher prediction error indicates the anomalies. However, the deep reconstruction-based approaches suffer from memorizing the low-level details rather than learning the full semantics of data [27]. Therefore, prediction-based approaches are in high demand. Due to their remarkable representation skills, Convolutional Neural Networks (CNNs) have been required for both reconstruction and prediction-based approaches. With the introduction of Generative Adversarial Networks (GANs) [17], adversarial training is employed in the VAD algorithms, resulting in improved reconstruction, prediction results, and performance gains. Despite their exceptional capability, CNN-based approaches are limited by the intrinsic locality of convolutional processes and fail to represent the long-term relations between the data. To address this, self-attention mechanisms are added into CNN-based algorithms [66], [54], significantly improving the capacity of CNN-based models to simulate complex features of input data.

Transformer, as a sequence-to-sequence model, has made substantial advances in the field of natural language processing (NLP)[50] [8], video prediction [29] and video captioning [30]. Vaswani et al. [50] initially introduced the transformer for machine translation, offering a viable alternative to the ex-

*Corresponding author

Email addresses: n.aslam921@gmail.com (Nazia Aslam), mahesh@iitp.ac.in (Maheshkumar H Kolekar)

isting NLP framework based on Recurrent Neural Networks (RNNs) and CNNs. Brown et al. [8] developed a Generative Pre-trained Transformer3 with 175 billion parameters. This framework is trained on a large-scale training dataset, capable of performing various NLP tasks, and does not require fine-tuning. Various research in computer vision has lately adopted the transformer framework due to the immense success of transformers in the field of NLP. For image classification, Dosovitskiy et al. [13] developed a vision transformer (ViT) that receives the input in a 16×16 patch of various tokens to the encoder of the transformer. Similarly, Arnab et al. [3] developed a video vision transformer (ViViT) that classifies the video data. ViViT takes the spatiotemporal tokens of input video data and then encodes it utilizing various transformer layers. Besides this, Chen et al. developed a TransUnet [10] that combines UNet and transformer networks and is utilized for medical image segmentation. Jiang et al. [24] presented a novel architecture, TransGAN, which incorporates a generator and a multi-scale discriminator tailored to be memory-friendly. Additionally, it is enhanced with a novel grid self-attention mechanism. In addition, Yuan et al. [64] developed a UNet-based transformer network incorporating GAN to detect anomalies. It has been observed that the transformer network performs better than the CNN networks if trained on large datasets. The receptive fields of the transformers are wider due to the multi-head self-attention layer and can capture the inherent property of spatiotemporal data. Therefore, inspired by the learning capability of the transformer network, we have utilized ViViT to learn the regular pattern of video data through adversarial training. A novel future frame prediction framework, TransGANomaly, has been designed to take the 3D token as input and predict the future frame with minimal error. The TransGANomaly amalgamates the GAN and ViViT networks and works together to detect anomalies. The generator of the GAN is the predictive ViViT network that tries to predict the future frame based on the spatiotemporal features of past frames. A CNN-based deep neural network has been designed to discriminate between the predicted frame and the original ground truth for adversarial learning. The summary and the contribution of our work are as follows:

1. A novel framework, TransGANomaly, is designed, which combines the properties of the video vision transformer and GAN network to learn the regular pattern of the video data through adversarial learning.
2. To keep the spatiotemporal features of the video data, N overlapping 3D tokens have been created by applying the convolution layer on the input video snippets. These N 3D tokens are given as input to the ViViT networks to learn the inherent spatiotemporal features and produce the future frame with minimal error.
3. Extensive experiments have been conducted on the three widely used datasets (UCSD Pedestrian, CUHK Avenue, and ShanghaiTech) and compared with the various state-of-the-art methods to show the efficacy of the proposed framework.

Acronyms	Nomenclature
VAD	Video Anomaly Detection
CNN	Convolutional Neural Network
GAN	Generative Adversarial Network
ConvLSTM	Convolutional long short-term memory
LSTM	Long short-term memory
NLP	Natural Language Processing
RNN	Recurrent Neural Network
ViT	Vision Transformer
ViViT	Video Vision Transformer
HOF	Histogram of Optical Flow
HOG	Histogram of Oriented Gradient
HMM	Hidden Markov Model
MPPCA	Mixture of Probabilistic Principle Component Analyzer
MDT	Mixture of Dynamic Textures
SF	Social Force
ConvAE	Convolutional Autoencoder
VAE	Variational Autoencoder
STAE	Spatiotemporal Autoencoder
MSE	Mean Square Error
ViViT	Video Vision Transformer
ViT	Vision Transformer
PSNR	Peak Signal to Noise Ratio
ROC	Receiver operating characteristics curve
AUC	Area under the curve

2. Related Work

Anomaly detection refers to identifying the events that deviate from the regular ones. Usual events can be treated as a positive class in a real-world scenario. In contrast, unusual events can be treated as a negative class, and negative classes are rare. Therefore, the annotations of the datasets are a tedious and time-consuming task. Based on the data annotations, anomaly detection techniques are divided into three categories: i) unsupervised methods: test the model using the unlabeled dataset, assuming that the aberrant pixels are infrequent compared to the regular one. ii) supervised methods: train the model on the normal and abnormal datasets with their annotations. iii) semi-supervised methods: trains the model only with the usual data with their labels. Broadly, anomaly detection techniques are classified into two groups: 1) Handcrafted feature-based anomaly detection and 2) Deep learning-based anomaly detection.

2.1. Handcrafted Features based Anomaly Detection

The handcrafted feature-based anomaly detection is categorized into two steps: feature extraction and anomaly detection. These features are extracted manually through a Histogram of Oriented Gradient (HOG), Histogram of Optical Flow (HOF) [1], texture [74], 3D gradient [52]. The commonly employed anomaly detection methods include clustering positive and negative classes [69], autoregressive integrated moving average model [57], reconstruction of the input data [2], and generation probability discrimination [53]. Also, some methods uti-

lized the motion trajectory of the objects to detect anomalous behaviour. In [12], Cosar et al. utilize the output trajectory of the moving object and analyze the pixel-level anomalous behaviour. The trajectory-based anomaly detection includes object detection, tracking, trajectory-based feature extraction, and detection. The advantage of using trajectory-based methods for anomaly detection is that it has highly computationally efficient and simple to implement. However, its performance deteriorates with the complex scene and occlusion since detecting and tracking the moving object during occlusion is difficult. To mitigate this, some statistical approaches like Hidden Markov Models (HMM)[21], the mixture of probabilistic principal component analyzers (MPPCA) on local optical flow patterns [25], Gaussian mixture model to the mixture of dynamic textures (MDT), and the social force model (SF) [39] were used to learn the typical patterns of the video data. Sparse coding is also a typical method for encoding the regular patterns of video data [34]. Mo et al. [40] proposed a trajectory-based jointly learned sparsity framework for detecting anomalies in videos with multiple objects. It creates a dictionary by sparsely representing the regular events. Anomalies are recognized when occurrences have a high reconstruction error. Even though many sparsity-based approaches are easy to implement, the computing cost of optimizing the sparse coefficients is expensive.

2.2. Deep Learning based Anomaly Detection

As discussed earlier, deep learning-based approaches are classified into two categories: Reconstruction-based and Prediction-based methods.

2.2.1. Reconstruction-based Anomaly Detection

The reconstruction-based method tries to recreate the input video data with minimal error. During inference, the higher reconstruction error signifies the occurrence of anomalies. Hasan et al. [18] developed a fully convolutional neural network to train the spatial features of normal video data. The regularity score of the frames is computed based on the reconstruction error during the inference stage. To jointly learn the spatial and temporal features of video data, Chong and Tay [11] and Luo et al. [35] combine ConvLSTM and ConvAE. The motion learning strategy helps to identify the anomalies and improves accuracy. Besides these techniques, the two-stream model was developed by Nguyen and Meunier [42], which uses ConAE with two branches to reconstruct the input frames and their corresponding optical flows. Furthermore, the reconstruction error of the pixel intensity and optical flow is used to compute the regularity score. Moreover, VAEs were employed by Fan et al. [14], and Li and Chang [14] to recreate the input frames. The distribution difference of the latent representations was utilized to calculate the regularity scores.

2.2.2. Prediction-based Anomaly Detection

The prediction-based method trains the model to predict the next sequence of input data based on the past sequences, and the prediction error is used to compute the anomaly. In [28], Li et al. developed a fast Fourier inception network for video

prediction that is composed of the occlusion inpainter and the spatiotemporal feature extractor. This network works on the occlusion scenario and can predict the future frame with minimal error. However, Zhong et al. [71] presented a motion matrix-based video prediction network. An autoencoder-based approach has been adopted to couple the appearance features with the motion matrix. In [38], Medel and Savakis proposed a conditional composite ConvLSTM-AE framework that employs two decoder networks to reconstruct the input frame and predict the future frame. To compute the regularity score, only the reconstruction error was considered. Similarly, Zhao et al. [70] constructed an STAE with two decoder branches; one is employed to recreate the input frame, and another is used to predict future frames. The regularity score is computed using both the reconstruction and prediction errors. Besides this, Liu et al. [32] proposed a UNet-based prediction framework. The method calculates the regularity score without reconstruction using only prediction error. In addition, various research includes reconstruction into prediction models. For example, Ye et al. [60] presented a Predictive Coding Network for anomaly detection that uses a ConvLSTM with predictive coding to predict future frames. The prediction error is described in terms of the reconstruction error. Lastly, the predicted frames are updated with the specified errors for better prediction accuracy. Also, Tang et al. [48] created a sequence connection between two U-Net blocks. The first block predicts the future frames, takes them as input, and tries to recreate them with minimal error. After that, the reconstructed predictions were utilised to compute the prediction error.

Table 1: Notation and their descriptions

Notation	Description
$\mathcal{F}_G$	Transformation function of the generator
$\mathcal{G}$	Generator
$\mathcal{D}$	Discriminator
X_i	i^{th} video snippet
$\hat{I}_{t+1}$	Predicted frame at time (t+1)
I_{t+1}	Original frame at time (t+1)
$\Phi(..)$	Transformation function to convert 3D patches into 1D vector
$\mathcal{E}_{pos}$	Postional embeddings
Y_{MSA}^l	Output of MSA at l^{th} layer
$\mathcal{L}_{int}$	Intensity loss
$\mathcal{L}_{grad}$	Gradient loss
$\mathcal{L}_G$	Generator loss
$\mathcal{L}_D$	Discriminator loss

3. Methodology

The notion of the proposed approach is that, given a sequence of t consecutive frames, $\{I_{t-T}, ..., I_{t-1}, I_t\}$, the model will predict the next frame $\hat{I}_{t+1}$ by exploiting the spatiotemporal features of the input data. In [32], Liu et al. demonstrated that the predictive approach improves the VAD accuracy compared to the reconstructive-based methods. Therefore, we have

developed a novel prediction-based VAD framework named TransGANomaly, shown in Figure 2, that predicts the future frame based on past frame information. TransGANomaly is a video vision transformer-based GAN network trained only on normal video data. The generator of the GAN is a ViViT network, and the discriminator is a CNN-based deep neural network that acts as a binary classifier. The generator of the proposed framework maps $\mathcal{F}_G : X_i \rightarrow X_{pred}$ by taking the 3D spatiotemporal tokens from the video data. Here, $X_i \in \{I_{t-T}, \dots, I_{t-1}, I_t\}$ is the current input frames and $X_{pred} \in \{\hat{I}_{t+1}\}$ is the predicted frame, and T is the temporal frame sequence. The goal of $\mathcal{F}_G$ is to generate X_{pred} taking X_i with minimal error. After that, the CNN-based discriminator receives the predicted frame and their corresponding ground truth to check between the real and fake data. To do this, we have employed adversarial learning between the frames that provide more discriminative results.

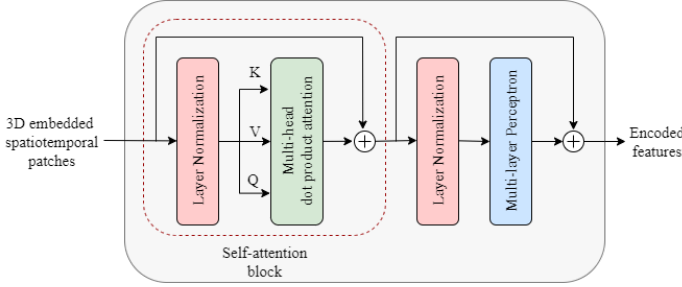


Figure 1: An illustrative diagram of the Transformer Encoder

Firstly, we have created n small video snippets $X_i \in \mathbb{R}^{T \times W \times H \times C}$, from the input video data by using the sliding window of size 10 and stride 1. Here T is the number of frames taken, which is 10 in our case, and W and H are the width and height of the frames, respectively. After that, Conv3D layers were used to construct the N 3D tokens consisting of spatiotemporal features of input data. Unlike ViViT [3], our 3D tokens are the overlapped 3D patches because we want to preserve all the temporal information of video snippets. These 3D tokens are linearised and then flattened into 1D vectors. Aftermath positional and token embeddings have been added to the 3D tokens and given to the transformer encoder for encoding. The encoded features received from the transformer layers are then fed to the deconvolution layer to predict the future frame $\hat{I}_{t+1}$. The predicted frame $\hat{I}_{t+1}$ and corresponding ground truth I_{t+1} are given to the discriminator that checks for real and fake data. To minimize the difference between $\hat{I}_{t+1}$ and I_{t+1} , intensity and gradient loss have been employed. Adversarial learning updates the internal parameters of the network, and for that, LS-GAN has preferred.

3.1. Generator: Predictive Video Vision Transformer

Following the work of video classification [3], we have developed a novel ViViT-based predictive video vision transformer (P-ViViT) network, shown in Figure 2, to detect video anomalies. P-ViViT can generate more discriminative features for anomaly detection if trained on a large dataset. Given an

input video snippet $X_i \in \mathbb{R}^{T \times W \times H \times C}$, where T is the number of frames in the temporal dimension, W and H is the width and height of the frames, and C is the channel, is divided into N overlapping patches (3D tokens) of size $t_p \times w_p \times h_p \times C$ and can be represented as $X_{ip} \in \mathbb{R}^{t_p \times w_p \times h_p \times C}$. These overlapping patches (3D tokens) are obtained by applying Conv3D layers with kernel size equal to patch size and stride $t_s \times w_s \times h_s$ on the video snippets X_i . To get the overlapping patches, the stride size is taken smaller than the kernel size. In our case, the kernel size is (4, 16, 16), and the stride size is (2, 4, 4). Therefore, for the input video snippet $X_i \in \mathbb{R}^{T \times W \times H \times C}$, the total number of overlapping patches N , of size $t_p \times w_p \times h_p$ can be obtained by the following equation:

$$N = N_t \times N_w \times N_h \\ = \lfloor \frac{T - t_p}{t_s} + 1 \rfloor \times \lfloor \frac{W - w_p}{w_s} + 1 \rfloor \times \lfloor \frac{H - h_p}{h_s} + 1 \rfloor \quad (1)$$

Here, N_t , N_w , N_h is the number of patches across temporal, width and height dimensions, respectively, and $\lfloor \dots \rfloor$ is the floor value. After that, these 3D tokens are linearised and flattened into a 1D vector of dimension D and can be represented by $X_{ip}^F \in \mathbb{R}^{N \times D}$. Therefore the n^{th} flattened patch is represented by the following equation:

$$X_{ip}^F(n) = \Phi(X_{ip}(n)) \quad (2)$$

Here Φ is the transformation function that convert 3D patches into 1D vector. After that positional and tokens embeddings are employed to incorporate the spatial information of the video snippets X_i . Then these embedded tokens, now called tensors, are passed through the L layers of the transformer for feature encoding. These tensors can be represented as:

$$Z_i = [\Phi(X_{ip}(1)) + \Phi(X_{ip}(2)) + \dots + \Phi(X_{ip}(N))] + \mathcal{E}_{pos} \quad (3)$$

Here, $Z_i \in \mathbb{R}^{N \times D}$ embedding sequence that is fed to the transformer network and $\mathcal{E}_{pos} \in \mathbb{R}^{N \times D}$ is the positional embedding.

Each layer of the transformer block consists of multi-head self attention [50], layer normalization [7] and multilayer perceptron units as shown in Figure 1. The MLP unit consist of two linear projection separated by GELU activation function. For l^{th} layer, the output of MSA unit and LN unit can be represented by the following equations:

$$Y_{MSA}^l = MSA(LN(Z_i^{l-1})) + Z_i^{l-1} \quad (4)$$

$$Z_i^l = MLP(LN(Y_{MSA}^l)) + Y_{MSA}^l \quad (5)$$

Here, Y_{MSA}^l is the output tensor for the l^{th} layer of MSA unit and Z_i^l is the output tensor of the transformer block. The tensor Z_i^l keeps the encoded spatiotemporal features of input data with their positional embeddings. Therefore, finally we have employed two deconvolution layer after the transformer block that works opposite to the convolution layer at the input side and predict the future frame $\hat{I}_{t+1}$. To minimize the prediction

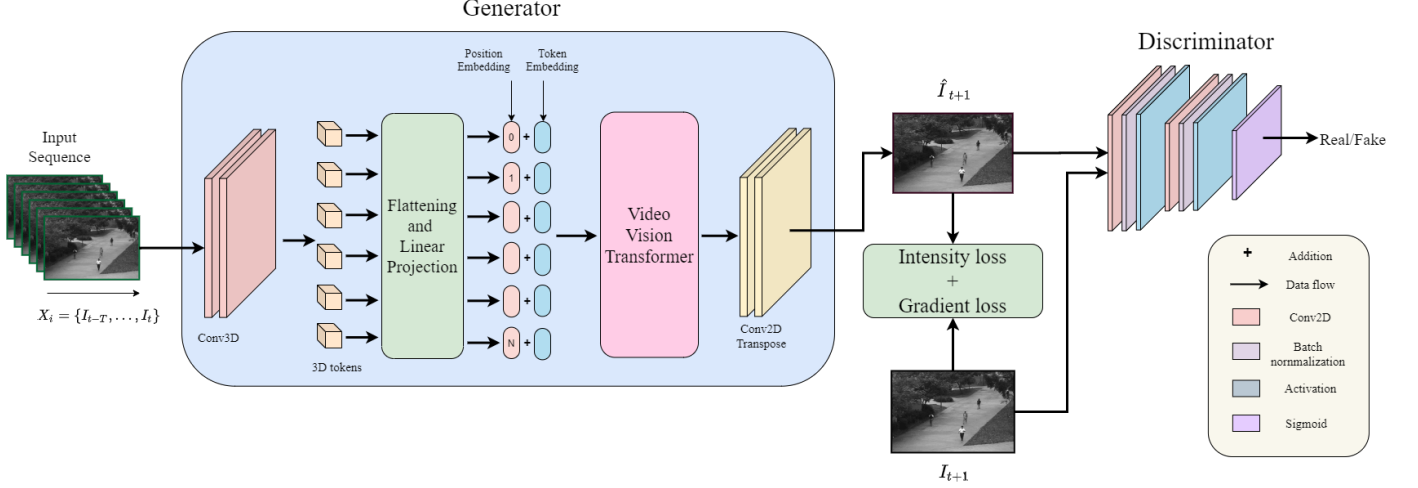


Figure 2: An illustrative diagram of the proposed framework

error between the predicted frame $\hat{I}_{t+1}$ and the original frame I_{t+1} , we have employed the intensity loss along with gradient loss. The intensity loss between I_{t+1} and $\hat{I}_{t+1}$ is given by the following equation:

$$\mathcal{L}_{int}(I_{t+1}, \hat{I}_{t+1}) = \frac{1}{WH} \|I_{t+1} - \hat{I}_{t+1}\|_2^2 \quad (6)$$

Here, W and H is the width and height of the image. The following equation describes the gradient loss and given as:

$$\mathcal{L}_{grad}(I_{t+1}, \hat{I}_{t+1}) = \frac{1}{WH} \sum_{(w,h)} \| |I_{w,h} - I_{w-1,h}| - |\hat{I}_{w,h} - \hat{I}_{w-1,h}| \|_1 + \| |I_{w,h} - I_{w,h-1}| - |\hat{I}_{w,h} - \hat{I}_{w,h-1}| \|_1 \quad (7)$$

Here w, h indicates the spatial location of the image pixel. The gradient loss guides the model in distinguishing between regular and irregular frames, ensuring that the model will generate only regular pixels of the objects with sharp edges. When predicting, irregular objects that have never appeared in the training data cannot be sharpened correctly. As a result, the anomalous objects have blurry edges, resulting in a large prediction error.

3.2. Discriminator: CNN-based Classifier

The discriminator of the proposed TransGANomaly is a CNN-based binary classifier that tries to distinguish between the predicted and the original frame by adversarial learning. The discriminator combines the convolution layer, batch normalization layer and activation function, as shown in Figure 2. The discriminator aims to receive the sequence frames and predicts the probability of whether the frames are generated by the generator or from the training data. The feeding of a succession of input data further allows $\mathcal{D}$ to employ temporal features, enabling $\mathcal{G}$ to produce temporally consistent frames, in contrast to the methodology suggested in [32], which imposes an optical flow constraint to balance the temporal consistency. In

contrast to the regular GAN, we have incorporated LS-GAN [37] because it generates higher-quality images with less error and is more stable during training [32]. As a result, LS-GAN is used as an optimization problem to penalize the predicted frame's data distribution to be very close to the ground truth. Finally, the adversarial loss is computed on the discriminator's final layer with activated feature maps.

3.3. Adversarial Training

The Generative adversarial network (GAN) was developed by Goodfellow et al. [17] and used a variety of computer vision applications like super-resolution, anomaly detection etc. A GAN comprises two frameworks, a generator $\mathcal{G}$ and a discriminator $\mathcal{D}$, that compete in a zero-sum game [51]. The generator's task is to collect samples from a latent space given by the training data and produce a realistic applicant. At the same time, the discriminator, generally a binary classifier, is engaged in determining whether the samples are actual or generated by the generator. We have employed this method to train our framework, which accepts a sequence of video frames as input and predicts the evolution of future frames.

3.3.1. Discriminator Training

The discriminator $\mathcal{D}$ tries to distinguish between the predicted $\hat{I}_{t+1}$ and original frame I_{t+1} . Therefore, the loss function for the discriminator $\mathcal{D}$ is given by the following equation:

$$\mathcal{L}_{\mathcal{D}}(I_{t+1}, \hat{I}_{t+1}) = \frac{1}{2m} \sum_{(w,h)} [\mathcal{D}(I_{t+1})_{w,h} - 1]^2 + \frac{1}{2m} \sum_{(w,h)} (\mathcal{D}(\hat{I}_{t+1})_{w,h} - 0)^2 \quad (8)$$

Here, w, h is the spatial location of the image and m is mini-batch size during training.

3.3.2. Generator Training

The generator's objective is to produce images closer to ground truth. During the training of generator $\mathcal{G}$, the weights of the discriminator $\mathcal{D}$ are fixed. $\mathcal{G}$ can be trained by minimizing the adversarial loss, defined as follows.

$$\mathcal{L}_{\mathcal{G}}(\hat{I}_{t+1}) = \frac{1}{2m} \sum_{(w,h)} [\mathcal{D}(\hat{I}_{t+1})_{w,h} - 1]^2 \quad (9)$$

3.4. Objective Function

To train discriminator $\mathcal{D}$, the cost function is defined as:

$$\begin{aligned} L_{\mathcal{D}} &= \mathcal{L}_{\mathcal{D}}(I_{t+1}, \hat{I}_{t+1}) \\ &= \frac{1}{2m} \sum_{(w,h)} [\mathcal{D}(I_{t+1})_{w,h} - 1]^2 \\ &\quad + \frac{1}{2m} \sum_{(w,h)} (\mathcal{D}[(\hat{I}_{t+1})_{w,h}] - 0)^2 \end{aligned} \quad (10)$$

To train the generator $\mathcal{G}$, the objective function is defined by the following equation:

$$\begin{aligned} L_{\mathcal{G}} &= \lambda_{int} \mathcal{L}_{int}(I_{t+1}, \hat{I}_{t+1}) \\ &\quad + \lambda_{grad} \mathcal{L}_{grad}(I_{t+1}, \hat{I}_{t+1}) \\ &\quad + \lambda_{\mathcal{G}} \mathcal{L}_{\mathcal{G}}(\hat{I}_{t+1}) \end{aligned} \quad (11)$$

Here, λ_{int} , λ_{grad} , and $\lambda_{\mathcal{G}}$ is the hyperparamets that is tuned during the training process. To train the proposed framework, firstly, we resized the frame into 256×256 , and then all the pixels were normalized between 0 and 1. Adam [26] based stochastic gradient descent is employed to optimize the weights of the framework. The minibatch is taken as 4, and the learning rates for the generator and discriminator were set as 0.0001 and 0.00001, respectively.

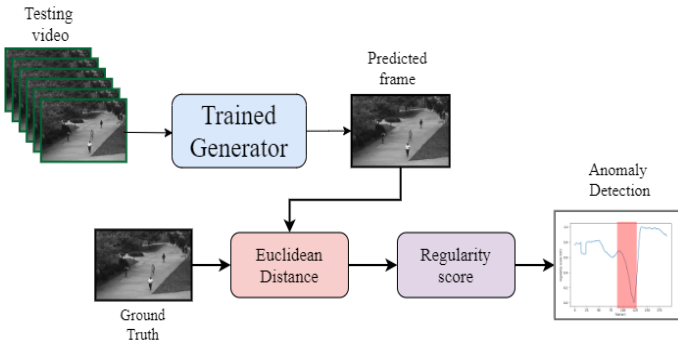


Figure 3: A diagrammatic illustration for the anomaly detection during testing phase.

3.5. Anomaly Detection

During the inference stage, the generator of the proposed framework is used to compute the regularity score of the testing data. Since the generator is responsible for predicting the next frame of the video data and hence can only predict the

regular pixel of the predicted frame, in contrast, the abnormal pixel will not be predicted. For a predicted frame $\hat{I}_{t+1}$ and its ground truth I_{t+1} , their difference will be used to compute the anomalous score. The PSNR is a popular metric for evaluating image quality in video anomaly detection and is given by the following equation:

$$PSNR(I_{t+1}, \hat{I}_{t+1}) = 10 \log_{10} \frac{[max_{\hat{I}_{t+1}}]^2}{\frac{1}{M} \|I_{t+1} - \hat{I}_{t+1}\|_2^2} \quad (12)$$

Here, $max_{\hat{I}_{t+1}}$ is the maximum value of the predicted frame $\hat{I}_{t+1}$ and M is the total number of pixel in the images. The highest value of PSNR indicates that the predicted frame closer to the normal frames, while the lowest value of PSNR indicates the unusual events happens in the video data. To compute the regularity score of the testing video, the PSNRs of the testing video is first normalized between 0 and 1 and then the regularity score of the $(t+1)^{th}$ frame is given by the following equation:

$$S_r(t+1) = \frac{PSNR(I_{t+1}, \hat{I}_{t+1}) - min(PSNR)}{max(PSNR) - min(PSNR)} \quad (13)$$

Here $max(PSNR)$ and $min(PSNR)$ indicates the maximum and minimum value of PSNR of the testing video.

4. Experiments

This section of the paper evaluates the efficacy of the proposed framework in detecting anomalies in the video sequence. We have evaluated the proposed framework, TransGANomlay, on the three most popular and extensively used datasets: the USCD Pedestrian dataset [31], recorded from pedestrian paths; the CUHK Avenue dataset [34], obtained from the mall's subway entry; and the ShanghaiTech dataset [36], gathered from various campus locations. As shown in Figure 5, we first evaluate TransGANomlay by computing the regularity score curve of the various testing video of the used dataset. Also, we have computed the localization of the anomalies within the frame, shown in Figure 6, which reflects that the proposed framework works well in detecting the anomalies. Also, the quantitative comparative analysis has been done on the various diversified state-of-the-art methods, as shown in Table 3, Table 4, and Table 5. Eight-core NVIDIA GeForce GTX 1080 GPUs have been used with a Tensorflow framework to conduct the experiment.

4.1. Dataset

We have employed the most common video anomaly datasets: UCSD Pedestrian [31], CUHK Avenue [34], and ShanghaiTech [36], to train and validate the performance of the proposed framework. These datasets are captured from a static camera at a fixed location. The training videos of the above-mentioned dataset consist only of normal events, while the testing video keeps both normal and abnormal events. The samples of some abnormal events in the above datasets are shown in Figure 4.

The UCSD Ped1 [31] dataset includes 34 training and 36 testing videos, each with 200 frames. The video clip includes

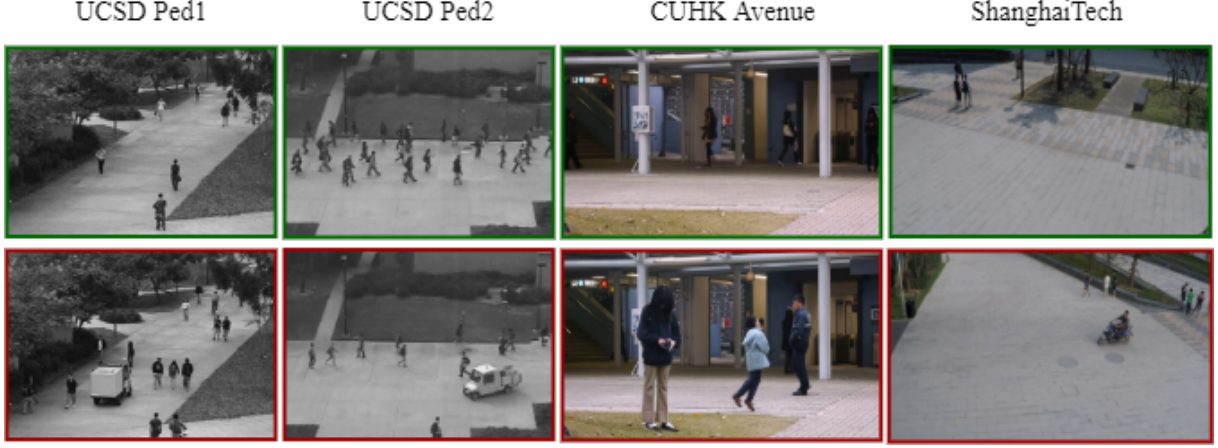


Figure 4: Samples of normal (upper) and abnormal (lower) frames of UCSD Ped1, UCSD Ped2, CUHK Avenue, and ShanghaiTech dataset.

Table 2: Specification comparison of the exercised dataset

Dataset	Total Frames/clips	Training Frames/clips	Testing Frames/clips	Anomalies	Background
UCSD Ped1	13,000/70	6,800/34	6,200/34	Bike, Skates, Vehicle	1
UCSD Ped2	4,560/28	2,550/16	2,010/12	Bike, Skates, Vehicle	1
CUHK Avenue	30,652/37	15,328/16	15,324/21	Running, Loitering, Hopping, Object throwing	1
ShanghaiTech	315,307/437	274,516/330	40,791/107	Bike, Skates, Vehicle, Sudden jump, Running etc.	13

people walking in groups towards and away from the camera’s viewpoint. In contrast, The **UCSD Ped2** dataset includes 16 training and 12 testing videos with different frame rates. People are seen walking straight from the camera viewpoint in the video clip. Anomalies in these datasets include people on bikes, skateboards, carts, and wheelchairs in the subway.

The **CUHK Avenue** [34] dataset possesses 16 training and 21 testing videos in less than a minute to almost two minutes. Persons strolling around the stairway and subway entrance are usual occurrences. However, unusual occurrences include: people rushing and moving in the wrong direction, loitering, tossing bags, hopping, and so on. The dataset’s challenges include camera movement and a few irregularities in the training samples.

The **ShanghaiTech** [36] dataset comprises 437 video clips with a frame resolution of 856×480 pixels and is the most extensive dataset for the unsupervised VAD. This dataset contains 318 training videos and 107 testing videos recorded from 13 diverse backgrounds settings. The videos are gathered by changing the camera angles and lighting variations, increasing the dataset’s diversity and bringing it closer to real-world settings. This dataset contains 130 real-world aberrant occurrences such as bicycles, skateboards, motorbikes, vehicles, fighting, chasing, and jumping.

4.2. Evaluation Metric

To check the efficacy of the proposed framework, we have employed the frame-based receiver operating characteristics

curve (ROC) and their corresponding area under the curve (AUC) as a performance parameter. The ROC is computed on a varying threshold between 0 and 1. The higher value of AUC is preferred to detect anomalies, while the lower AUC indicates the model has trouble detecting anomalies.

4.3. Implementation Details

Before feeding the input video data to the proposed framework, some pre-processing steps were performed. Each input frame is firstly converted to grayscale format and resized into 256×256 pixels. To maintain the same scale, each pixel of video data is adjusted between 0 and 1, then normalized by subtracting each frame from its global mean.

To compute the 3D tokens of size $4 \times 16 \times 16$ from input video cuboid X_i , two Con3D layers of filters 256 and 128 with kernel size (4, 16, 16) have been employed. After that, positional embeddings were applied to all 128 3D tokens. For layer normalization of the ViViT network, $1e-6$ has opted as an epsilon value. In contrast, for the MLP layer of the ViViT network, two dense layers with dimensions (512×1) and (128×1) with *Gelu* activation function have been taken. Eight layers of multi-head self-attention and eight layers of transformer blocks with a hidden dimension of 128 are considered for experimentation. Further, two Conv2D layers of size 128 and 256 of filter size 7×7 and 11×11 have been employed to predict the future frame from the input video cuboid. To train the whole model, Adam [26] optimizer with learning rate $2e-4$, decay $1e-5$ and epsilon $1e-6$ is considered for the generator $\mathcal{G}$, whereas

2e-5, decay 1e-6 and epsilon 1e-6 are considered for the discriminator $\mathcal{D}$. All the kernels weight for the Conv layers were initialized by the zero mean gaussian distribution with a standard deviation of 0.2. *Relu* activation function is used for the generator network, whereas *LRelu* with a slope of 0.3 for the intermediate state of the discriminator. At the end of the discriminator, a *sigmoid* activation function later classifies the samples with the label ‘Real’ and ‘Fake’. We train TransGANomaly with the above datasets either up to 200 epochs or until the validation loss becomes constant or minimum for five consecutive epochs. A batch size of four is considered for the entire experiment.

4.4. Quantitative Analysis

In this section, we have compared our proposed framework with the numerous recent and state-of-the-art methods. These methods are reconstruction-based, future prediction-based and hybrid methods. Table 3, Table 4, and Table 5 illustrate the comparison results for the UCSD Pedestrian, CUHK Avenue, and ShanghaiTech datasets in terms of Area under the curve (AUC). The comparison results are taken directly from their respective published papers. The performance of TransGANomaly for the UCSD Ped1, CUHK Avenue, and ShanghaiTech datasets is better among all the methods. For the UCSD Ped2 dataset, TransGANomaly is inferior, about 1.2% in terms of AUC than HF^2 -VAD [33]. But the performance of TransGANomaly is better in CUHK Avenue and ShanghaiTech datasets, shown in Table 4 and Table 5, respectively, which reflects that the proposed framework is able to learn the various diversified background of the dataset and able to perform well.

Table 3: Comparative analysis of the proposed framework in terms of AUC with the different state-of-the-art methods on UCSD Dataset. The best result is in bold.

Algorithm	Reference	Ped1 AUC(%)	Ped2 AUC(%)
MPPCA [56]	CVPR2010	66.8	69.3
MPPCA+SF [56]	CVPR2010	74.2	61.3
ConvAE [18]	CVPR2016	81.0	90.0
Stacked RNN [36]	ICCV2017	—	92.2
Spatiotemporal AE [11]	ISSN2017	89.9	87.4
FFP [32]	CVPR2018	—	95.4
Yan et al. [58]	IEEE TCSDS2018	75.0	91.3
ConvLSTM-AE [35]	ICME2017	75.5	88.1
Hybrid AE [53]	Optik2018	90.5	88.9
GANs [44]	ICIP2017	—	88.4
TAC-Net [23]	IEEE TII2021	—	98.1
ISTL [41]	IEEE TII2019	75.2	91.1
Wang et al. [55]	ICSP2018	86.7	92.5
Zhou et al. [73]	IEEE TCSVT2019	83.9	96.0
Zhang et al. [68]	IEEE TCSVT2020	—	95.4
AnomalyNet [72]	IEEE TIFS2019	83.5	94.9
Chen et al. [9]	IVC2020	89.0	96.6
Ada-Net [46]	IEEE TM2019	90.4	90.7
BR-GAN [59]	IEEE Access2021	84.7	97.6
HF^2 -VAD [33]	ICCV2021	—	99.3
SSAGAN [22]	IEEE TNNLS	84.2	96.9
Proposed framework		92.3	98.1

Table 4: Comparative analysis of the proposed framework in terms of AUC with the different state-of-the-art methods on Avenue Dataset. The best result is in bold.

Algorithm	Reference	AUC (%)
Conv-AE [18]	CVPR2016	70.2
Stacked RNN [36]	ICCV2017	81.7
Spatiotemporal AE [11]	ISSN2017	80.3
ConvLSTM-AE [35]	ICME2017	77.0
Lu et al. [34]	ICCV2013	80.9
Chen et al. [9]	IVC2020	89.0
MNAD-R [43]	CVPR2020	88.5
Unmasking [49]	ICCV2017	80.6
AnoPCN [60]	ICM2019	86.2
Yan et al. [58]	IEEE TCDS2018	79.6
Zhou et al. [73]	IEEE TCSVT2019	86.0
Multispace [68]	IEEE TCSVT2020	86.8
FFP [32]	CVPR2018	84.9
ISTL [41]	IEEE TII2019	76.8
Wang et al. [55]	ICSP2018	79.1
MemAE [16]	ICCV2019	83.3
BR-GAN [59]	IEEE Access2021	88.6
HF^2 -VAD [33]	ICCV2021	91.1
SSAGAN [22]	IEEE TNNLS2022	88.8
TAC-Net [23]	IEEE TII2021	88.8
Proposed framework		90.2

Table 5: Comparative analysis of the proposed framework in terms of AUC with the different state-of-the-art methods on ShanghaiTech Dataset. The best result is in bold.

Algorithm	Reference	AUC (%)
MemAE [16]	ICCV2019	71.2
MNAD-R [43]	CVPR2020	69.8
FFP [32]	CVPR2018	72.8
MNAD-P [43]	CVPR2020	70.5
VEC [61]	ICM2020	74.8
AnoPCN [60]	ICM2019	73.6
HF^2 -VAD [33]	ICCV2021	76.2
Multispace [68]	IEEE TCSVT 2020	73.3
CT-D2GAN [15]	ICM2021	77.7
TAC-Net [23]	IEEE TII2021	77.2
Zhang et al. [67]	ICIP2019	83.5
Zaheer et al. [65]	IEEE SPL2020	84.1
Proposed framework		85.3

4.5. Qualitative Analysis

4.5.1. Analysis of regularity score curve

To visualize the frame-level abnormalities in the testing videos, we have plotted the regularity score curve versus the frame number on the above-used dataset, as shown in Figure 5. The proposed TransGANomaly is trained only on the normal video data, so the regularity score of the regular events is high. For TransGANomaly, it is challenging to predict the abnormal pixel of the video data; therefore, there is a sudden decline in the curve when an abnormal event comes into the camera’s viewpoint. We have generated the curve between the regularity score and frame number by utilizing Equation 13. We can infer from Figure 5 that the proposed framework retains a high regularity score during regular occurrences. However, the regularity score drops sharply when an unusual pixel enters the frame.

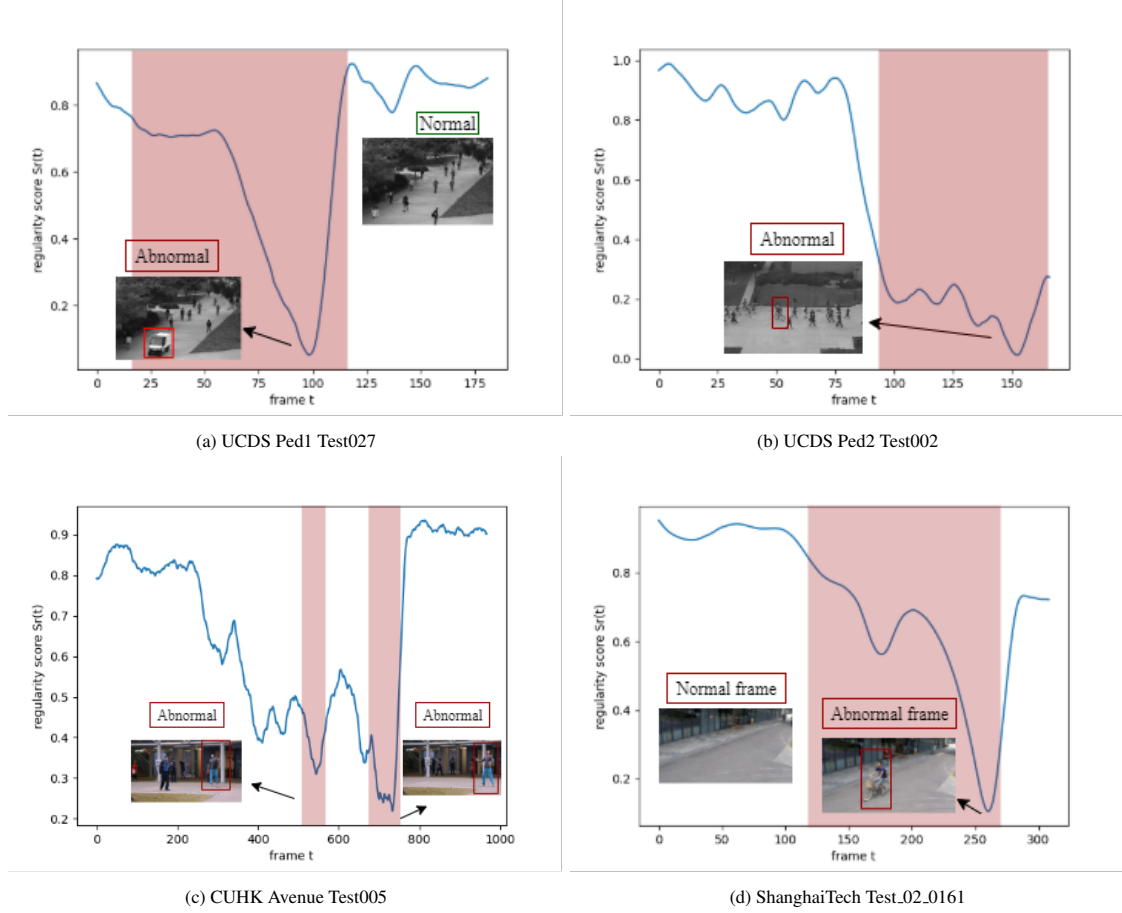


Figure 5: Regularity score curve for testing videos of the different datasets. The red color region indicates the ground truth. Best viewed in colors.

4.5.2. Anomaly localization

The proposed framework is trained only on the normal video data; therefore unable to predict the irregular pixels. Hence, the predicted frame by the TransGANomaly with normal pixels will be predicted with minimal error. In contrast, the proposed framework does not predict abnormal pixels and leads to a blurry image. To localize anomalies inside the frame (spatial anomaly localization), we have subtracted the predicted frame with their corresponding ground truth and given by the following equation:

$$\text{sub}(I_{t+1}) = |I_{t+1} - \hat{I}_{t+1}| \quad (14)$$

After the thresholding of the subtracted frame, the heatmap is drawn on the image, showing the anomalies with a colour. The cart is an anomaly for the UCSD Pedestrian dataset; for the CUHK Avenue dataset, the boy is hopping and considered an anomaly, while in the ShanghaiTech dataset, there is a biker and viewed as an anomaly. Figure 6 illustrates the anomaly localization of the proposed framework on the four datasets, which reflects the efficacy of the proposed framework. Hence, the anomalies are identified within the frame of the video data.

4.6. Computational Complexity

The computational complexity of the proposed TransGANomaly depends on the computation of the self-

attention mechanism of the transformer network and the training of the GAN. The generator of the TransGANomaly is a video vision transformer network, while the discriminator is a CNN-based classifier.

The self-attention mechanism in the transformer is a key component. For a single attention head, the time complexity is typically $O(N^2 \times D)$, where N is the sequence length (number of frames in a video) and D is the dimension of the input features. For H attention heads and L layers in the network, the overall time complexity becomes $O(L \times H \times N^2 \times D)$. Since the discriminator of the model is a CNN-based classifier, therefore, the computation of the discriminator is given by the following equation:

$$T_c \sim O(F_h F_w \cdot K_h K_w \cdot C_K C_F) \quad (15)$$

Here, F_h, F_w are the height and width of feature maps, K_h, K_w are the height and width of kernels, and C_K, C_F are the channels of kernels and feature maps respectively.

4.7. Ablation analysis

We have investigated the effect of the overlapping and non-overlapping patches on the proposed TransGANomaly for detecting anomalies on the UCSD Pedestrian, CUHK Avenue, and ShanghaiTech datasets, shown in Table 6. The proposed

Table 6: AUC of the TransGANomaly on a different settings of the transformer network using an non-overlapping 3D STC tokens.

	UCSD Ped1	UCSD Ped2	CUHK Avenue	ShanghaiTech
Non-Overlapped STC tokens	88.7%	93.6%	86.1%	80.3%
Overlapped STC tokens	92.3%	98.1%	90.2%	85.3%



(a) UCSD Ped1 Test027



(b) UCSD Ped2 Test004



(c) CUHK Avenue Test021



(d) ShanghaiTech Test 01_0161

Figure 6: Anomaly localization within the frame on the aforementioned dataset. left: original frame, right: heatmap of anomalies. Best viewed in colors.

TransGANomaly takes input as 3D tokens created from the input video cuboid by employing the CNN layer. These 3D tokens consist of spatial and temporal features of the input video data. Arnab et al.[3] utilize the non-overlapping patches as input for the video vision transformer (ViViT). These non-overlapping patches contain only the video data’s spatial features and neglect the temporal relation between the tokens. Apart from this, our framework uses overlapping patches and constructs 3D spatiotemporal tokens which consist of the spatio-temporal features of the video data. After conducting the experiments using non-overlapping and overlapping patches,

Table 7: AUC of the proposed framework with different loss function

Dataset	Intensity loss	Gradient loss	Intensity + Gradient loss
UCSD Ped1	90.2%	89.3%	92.3%
UCSD Ped2	93.8%	92.9%	98.1%
CUHK Avenue	86.3%	87.1%	90.2%
ShanghaiTech	80.1%	82.1%	85.3%

we can say that overlapping patches gave better results than non-overlapping patches. The AUC of the proposed framework on UCSD Ped1, UCSD Ped2, CUHK Avenue and ShanghaiTech datasets has improved by 3.6%, 4.5%, 4.1%, and 5% after utilizing the overlapping patches. Therefore, using overlapping patches to construct 3D spatiotemporal tokens for time series data is a good choice.

We have also conducted the experiments by varying the different losses, as shown in Table 7. We have utilized the combination of intensity and gradient loss for training the proposed framework. Intensity loss is the l_2 distance between the predicted and the original frame. After training the algorithm, using the intensity loss, the edges of the objects in the video data are blurry since it is a kind of averaging operation. To overcome this, we added the gradient loss with intensity loss. Therefore the addition of both the loss function gives the improved results as shown in Table 7. By employing the combination of intensity and gradient loss, the AUC of the proposed framework in UCSD Ped1, UCSD Ped2, CUHK Avenue, and ShanghaiTech datasets are enhanced by 2.1%, 4.3%, 3.9%, and 5.2%, respectively.

5. Conclusion

This paper aims to develop an optimized algorithm that can efficiently learn the regular motion patterns of video data to find abnormal events. We propose a novel framework, TransGANomaly, amalgamating the video vision transformer (ViViT) and the GAN network to detect anomalies. The generator of the TransGANomaly is a predictive ViViT network that aims to produce the future frame of the video data based on the past sequence. To learn the spatiotemporal features of the video snippets, N overlapping 3D tokens have been created and given as input to the ViViT network to produce the future frame. The discriminator of the GAN is a CNN-based deep neural network that receives the predicted frame by the ViViT network and their corresponding ground truth to discriminate between the real and fake data. Extensive experiments have been conducted on the UCSD Pedestrian, CUHK Avenue, and ShanghaiTech datasets to compute the qualitative and quantitative analysis. The qualitative and quantitative analyses have demonstrated that the proposed framework can detect frame-level anomalies and local-

ize the anomalies within the frames. **The model amalgamates the ViViT network with GAN; therefore, the training data must be extensive, and computing resources are required to achieve better results. Also, this network is not capable to handle the occlusion scnerios or the objects which can be anomaly. In future, we will extend this work to more challenging and real-world datasets which handle the occlusion scenario for fast and real-time anomaly detection.**

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